

arXiv:2504.03553

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM agents when to think fast, reflect deeply, or
seek external knowledge — through situational self-
awareness.

Paper: arxiv.org/abs/2504.03553

Current Agent Training Uses "Flood Irrigation" — Indiscriminate Knowledge Injection Hurts Efficiency

ReAct

Trajectory Imitation

Directly mimics gold trajectories without situational awareness of when guidance is actually needed.

Reflexion

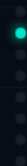
Trial-and-Error

Blindly injects external feedback after every failure, regardless of whether reflection alone could suffice.

KnowAgent / WKM

Knowledge-Augmented

Dumps domain knowledge at every step — a "flood irrigation" approach that overwhelms the agent.



Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking

At each decision step,
heuristic criterion
classifies:

Fast Thinking

Predicted action matches gold action directly.

<FAST>

If initial action is
incorrect:

Slow Thinking

Reflection fixes the incorrect action.

<REFLECT>

Knowledgeable Thinking

Even after reflection, external knowledge is needed.

<KNOWLEDGE>

KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

Step 1

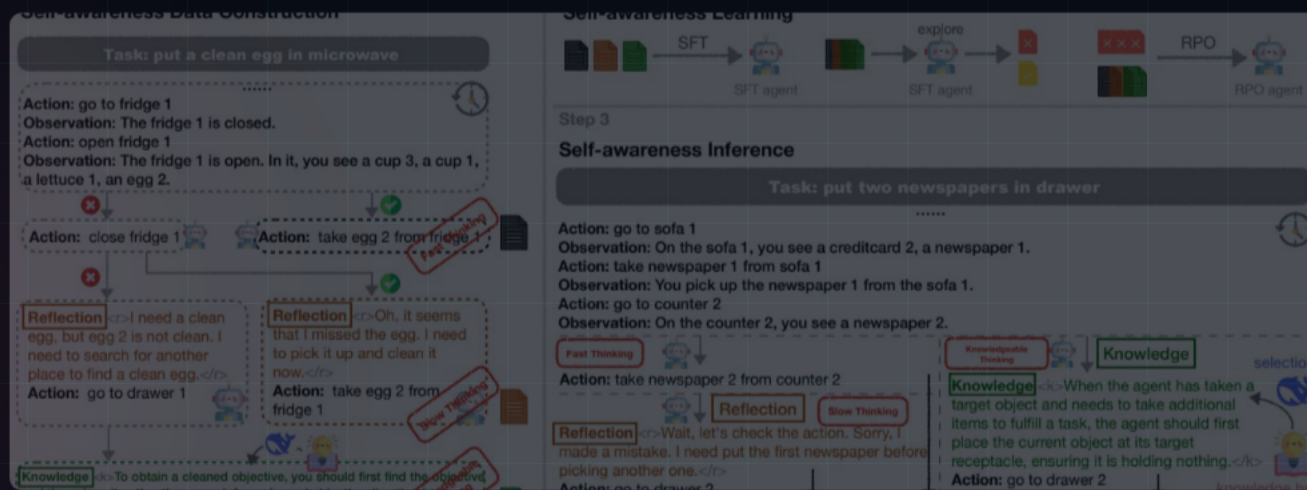
Self-Awareness Data Construction

Agent self-explores environments. Heuristic criterion classifies each step into fast / slow / knowledgeable. Special tokens are inserted to mark each situation type.

Step 2

Two-Stage Training

First, supervised fine-tuning (SFT) teaches initial self-awareness patterns. Then, Reward-based Policy Optimization (RPO) refines the agent's ability to generate appropriate special tokens.



KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Task: "put a hot mug in cabinet"

Heating the Mug

OpenAI-01 Incorrectly tries to take an egg from the microwave after opening it.

KnowSelf Uses a <REFLECT> token to realize it should heat the mug using the open microwave.

Task: "put a saltshaker in drawer"

Placing the Object

DeepSeek-R1 Reflects that the drawer is open and tries to close it first — which fails.

KnowSelf Applies a <KNOWLEDGE> token noting the agent should place the object without removing unrelated items.

Action: go to microwave 1

Observation: The microwave 1 is closed.

Action: open microwave 1

Observation: You open the microwave 1. In it, you see an egg 1.

[Knowledge]<k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k>

Action: take egg 1 from microwave 1

Observation: Nothing happens.

Action: go to microwave 1

Observation: The microwave 1 is closed.

Action: open microwave 1

Observation: You open the microwave 1. In it, you see an egg 1.

Action: put mug 1 in/on microwave 1

[Reflection]<r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r>

Action: heat mug 1 with microwave 1

.....

.....

(b) Task: put a saltshaker in drawer

.....

Action: take saltshaker 1 from countertop 3

Observation: You pick up the saltshaker 1 from the countertop 3.

Action: go to drawer 1

Observation: The drawer 1 is open. In it, you see a fork 1.

Action: put saltshaker 1 in/on drawer 1

[Reflection]<r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r>

.....

Action: take saltshaker 1 from countertop 3

Observation: You pick up the saltshaker 1 from the countertop 3.

Action: go to drawer 1

Observation: The drawer 1 is closed.

Action: open drawer 1

Observation: You open the drawer 1. In it, you see a fork 1.

[Knowledge]<k>When the agent encounters a situation where receptacles

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage



Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

- 01 Indiscriminate knowledge injection is **wasteful and suboptimal** — “flood irrigation” hurts both efficiency and performance.
- 02 Agents can learn **situational self-awareness** via lightweight special-token mechanisms with heuristic labeling — no expensive human annotation needed.
- 03 KnowSelf with Llama-8B achieves **84.33%** overall, outperforming GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%).

